Tutorial 10 Analysis

Problem 28

(a) Let $f \in L^s$. Then $\int_\Omega |f|^s \, d\mu < \infty$.

$p = \frac{s}{r}$, $\frac{1}{p} = \frac{r}{s}$

Let $\tilde{f} = |f|^r$

$\tilde{g} = 1$

for q

so $\frac{1}{q} = 1 - \frac{r}{s} = \frac{s-r}{s}$

so $q = \frac{s}{s-r}$

Since $(\tilde{f})^{s/r} = f^s$, it follows $\tilde{f} \in L^{s/r}$.

Since $\mu(\Omega) < \infty$, it follows $\tilde{g} \in L^{s/(s-r)}$

By Hölder's Inequality, $\tilde{f}\tilde{g} = f^r \in L^1 \Leftrightarrow f \in L^r$, and

$$\|\tilde{f}\tilde{g}\|_1 \leq \|\tilde{f}\|_{s/r} \|g\|_{1-r/s}$$

$$\Leftrightarrow \int_\Omega |f|^r \, d\mu \leq \left(\int_\Omega |f|^s \, d\mu\right)^{r/s} \cdot \left(\int_\Omega 1 \, d\mu\right)^{\frac{s-r}{s}}$$

$$\Leftrightarrow \left(\int_\Omega |f|^r \, d\mu\right)^{1/r} \leq \left(\int_\Omega |f|^s \, d\mu\right)^{1/s} (\mu(\Omega))^{\frac{s-r}{sr} = \frac{1}{r} - \frac{1}{s}}$$

(b) First note that $\frac{1}{r} = \frac{1}{p} + \frac{1}{q} \Leftrightarrow r = \frac{pq}{p+q}$

$\Leftrightarrow pr + qr = pq$

$\Leftrightarrow \frac{r}{q} + \frac{r}{p} = 1$

So let $\tilde{p} = \frac{p}{r}$, $\tilde{q} = \frac{q}{r}$, then $\frac{1}{\tilde{p}} + \frac{1}{\tilde{q}} = 1$.

Let $\tilde{f} = f^r$

$\tilde{g} = g^r$

Then $f \in L^p \Rightarrow \tilde{f} \in L^{\tilde{p}}$

$g \in L^q \Rightarrow \tilde{g} \in L^{\tilde{q}}$

By Hölder's Inequality, $\tilde{f}\tilde{g} \in L^1 \Leftrightarrow fg \in L^r$ and

$$\|\tilde{f}\tilde{g}\|_1 \leq \|\tilde{f}\|_{\tilde{p}} \|\tilde{g}\|_{\tilde{q}}$$

$$\Leftrightarrow \int_\Omega |fg|^r \, d\mu \leq \left(\int_\Omega |f|^p \, d\mu\right)^{r/p} \left(\int_\Omega |g|^q \, d\mu\right)^{r/q}$$

$\Leftrightarrow \|fg\|_r \leq \|f\|_p \|g\|_q$

Problem 29

(a) Let E be a positive set and $A \subseteq B \subseteq E$.
Then $E \setminus B \subseteq E$, and so $\mu(E\setminus B) \geq 0$, thus $\mu(E) = \mu(B) + \mu(E \setminus B) \geq \mu(B)$.
Similarly, ~~from $E \setminus A \subseteq E$ and~~ $B \setminus A \subseteq E$, we get $\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A) \geq 0$
So $0 \leq \mu(A) \leq \mu(B) \leq \mu(E)$

For a negative set, $F \supseteq B \supseteq A$, $F \setminus B \subseteq F$
So $\mu(F \setminus B) \leq 0$ so $\mu(F) = \mu(F \setminus B) + \mu(B)$
$\leq \mu(B)$.
Thus, doing the same as above, we get $\mu(F) \leq \mu(B) \leq \mu(A) \leq 0$

(b) Let $B \subseteq E$ be measurable and E positive. For all $A \in \mathcal{A}$, $A \subseteq B$, since $A \subseteq E$ $\mu(A) \geq 0$. So B is positive
Let $B \subseteq F$ be measurable and F negative.
For all $A \in \mathcal{A}$, $A \subseteq B \Rightarrow A \subseteq E \Rightarrow \mu(A) \leq 0 \Rightarrow B$ is negative

(c) Let $\{A_n\}_{n \in \mathbb{N}}$ be a sequence of positive sets, and $E \subseteq \bigcup_{n \in \mathbb{N}} A_n$, $E \in \mathcal{A}$
Let $B_n := A_n \setminus (\bigcup_{m=1}^{n-1} A_m)$. Then $B_n \subseteq A_n \Rightarrow B_n$ is positive, and $E \subseteq (\bigcup_{n=1}^{\infty} B_n)$.
So we show $\bigcup_{n=1}^{\infty} B_n$ is positive and then $\mu(E) \geq 0$.
$\mu(\bigcup_{n=1}^{\infty} B_n) = \sum_{n=1}^{\infty} \mu(B_n) \geq 0$. The countable union of negative sets is the same, but with $\sum_{n=1}^{\infty} \mu(B_n) \leq 0$.

Problem 30

(a) $\nu \ll \mu \iff \forall A \in \mathcal{A},\ \mu(A) = 0 \Rightarrow \nu(A) = 0$

Now $\nu(A) = 0 \iff \nu^+(A) - \nu^-(A) = 0$

$\iff \nu^+(A) = \nu^-(A)$

But $\nu^+(A) > 0 \iff \nu^-(A) = 0$, so $\nu^+(A) = \nu^-(A)$

$\iff \nu^+(A) = \nu^-(A) = 0$.

(b) Since $\nu(A) \leq \mu(A)$, we have that

$\mu(A) = 0 \Rightarrow \nu(A) = 0 \quad \Rightarrow \nu \ll \mu$

(ν is a finite measure $\Rightarrow$ signed measure)

By the Radon-Nikodym theorem, there exists some $f: \Omega \to \overline{\mathbb{R}}_+$ with $f = \frac{d\nu}{d\mu}$.

We have that

$\nu(A) = \int_A f \, d\mu$, and $f \geq 0$ since ν is a measure.

But $\nu(A) \leq \mu(A) \Rightarrow \int_A f \, d\mu \leq \int_A 1 \, d\mu$

Let $E = \{f > 1\} \in \mathcal{A}$.

Then $\nu(E) \Rightarrow \int_E f \, d\mu \geq \int_E 1 \, d\mu = \mu(E)$

So $\nu(E) = \mu(E) \iff \int_E (f-1) \, d\mu = 0$. But then $(f-1)\chi_E = 0$ μ-a.e., and $f > 1$ on $E \Rightarrow \mu(E) = 0$.

So $f \leq 1$ μ-a.e.